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(54) **ADJUSTABLE EQUIPMENT STORAGE
RACK FOR A BOAT**

(52) **U.S. Cl.**

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(57)

ABSTRACT

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8, 2024.

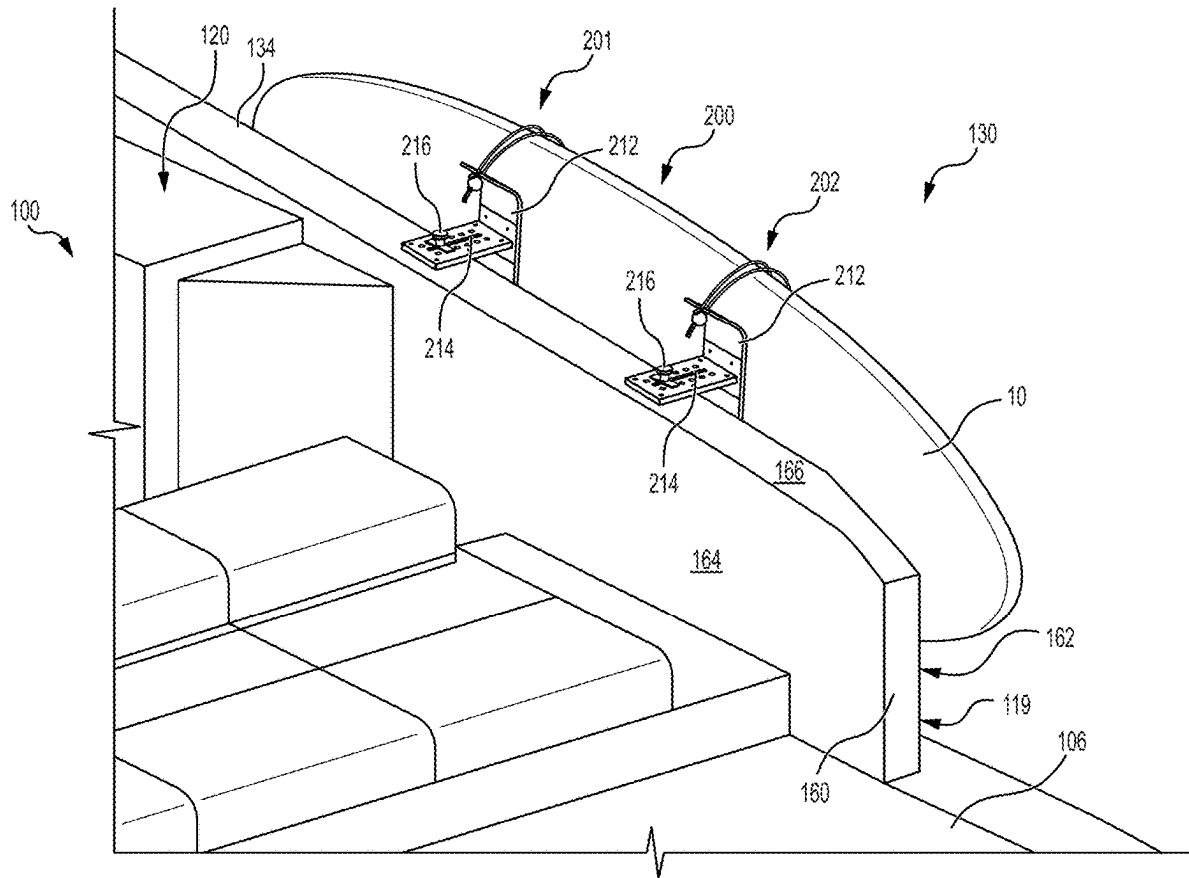
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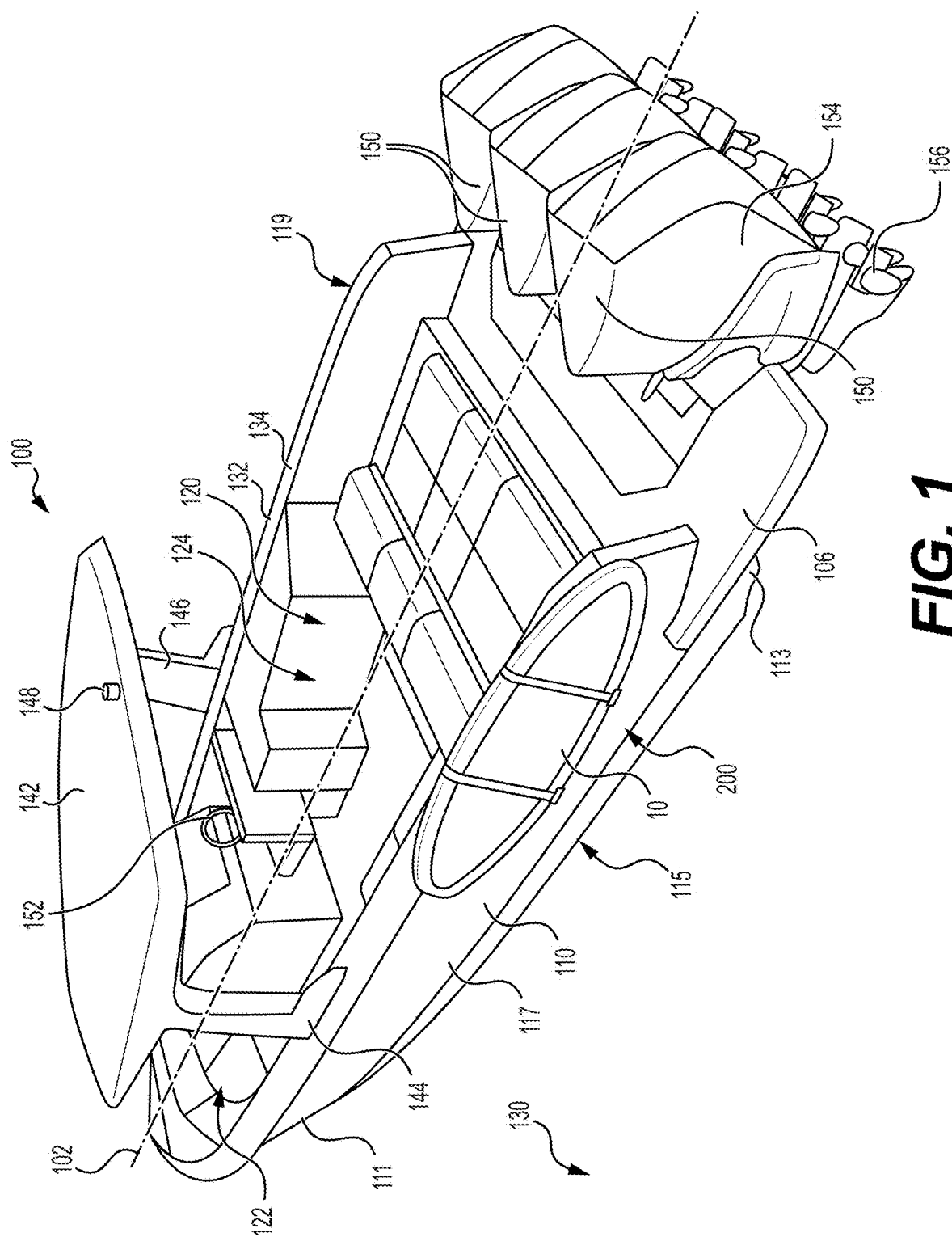
(51) **Int. Cl.**

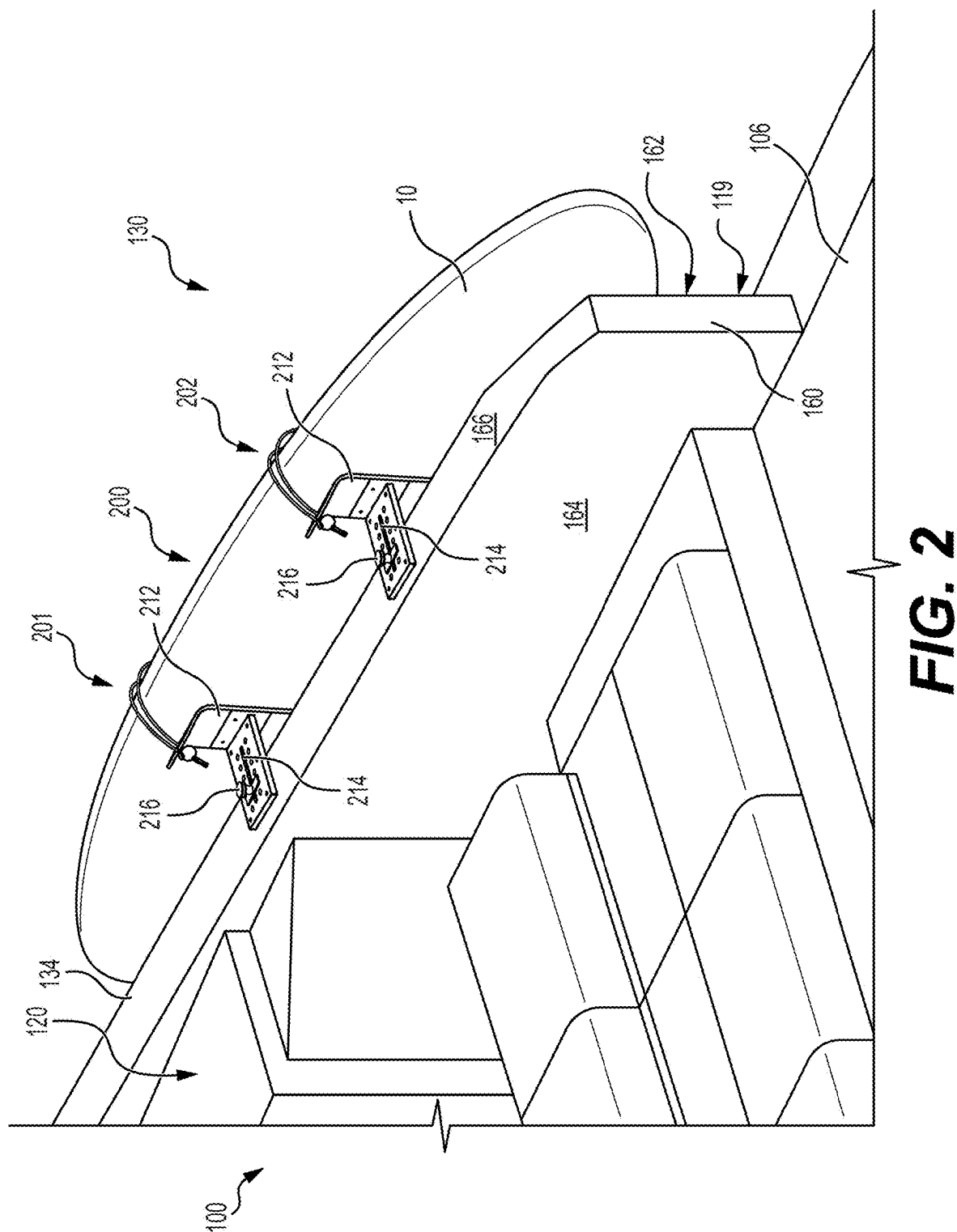
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(2020.01)

An equipment storage rack for a boat can store a piece of equipment on the exterior of the boat and can be adjustable. For example, the equipment storage rack can be attached to an upper perimeter surface of a barrier of the boat and can be adjustable for different widths of the upper perimeter surface. The adjustable equipment storage rack includes a cradle for supporting a piece of equipment, and a mounting bracket connected to the cradle to support the cradle. The mounting bracket has a plurality of engagement positions for a fastener to engage with the mounting bracket and to secure the mounting bracket to the boat at a fastening position. The cradle is a different distance from the fastening position at each engagement position of the plurality of engagement positions.







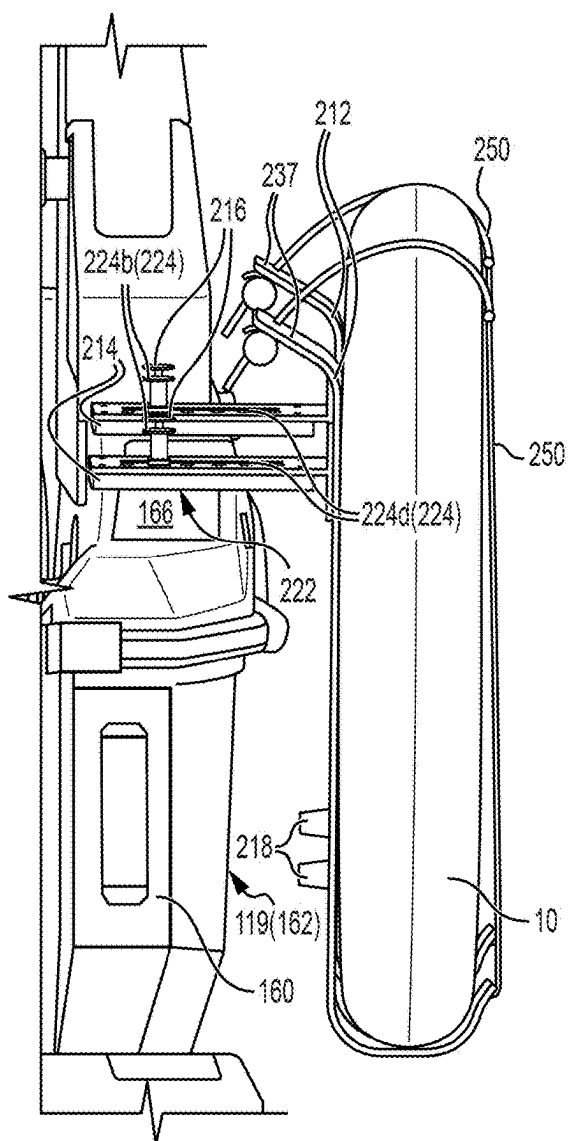


FIG. 3A

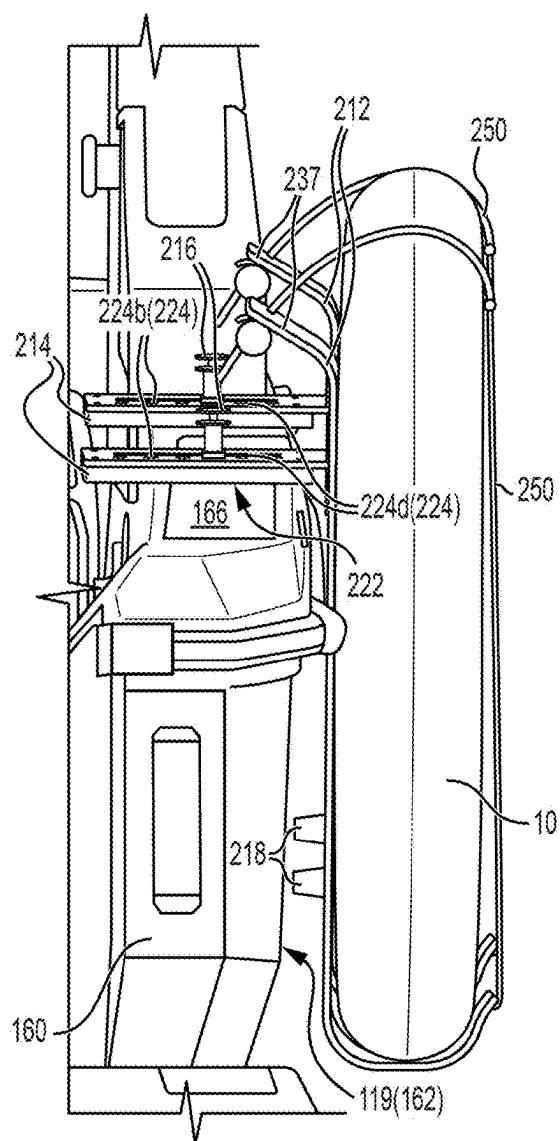


FIG. 3B

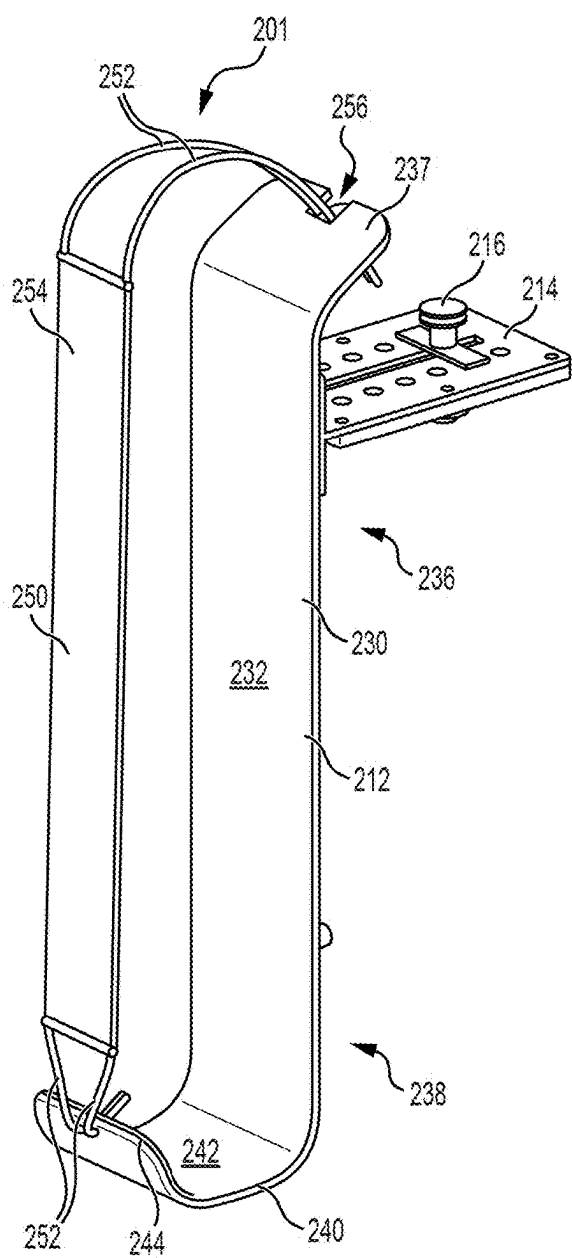


FIG. 4A

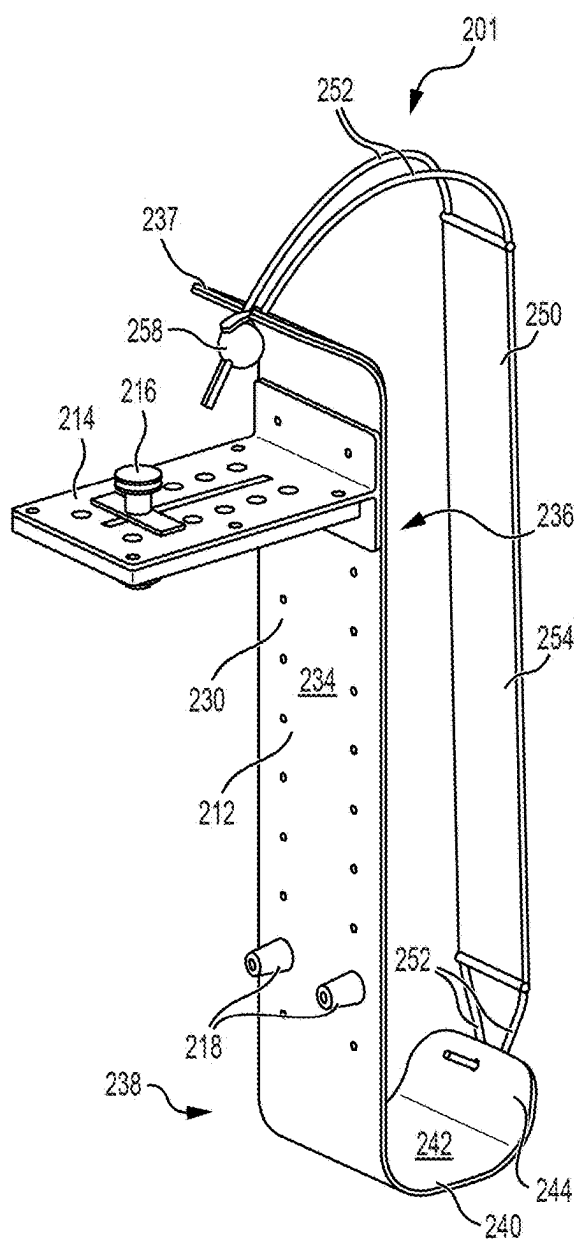
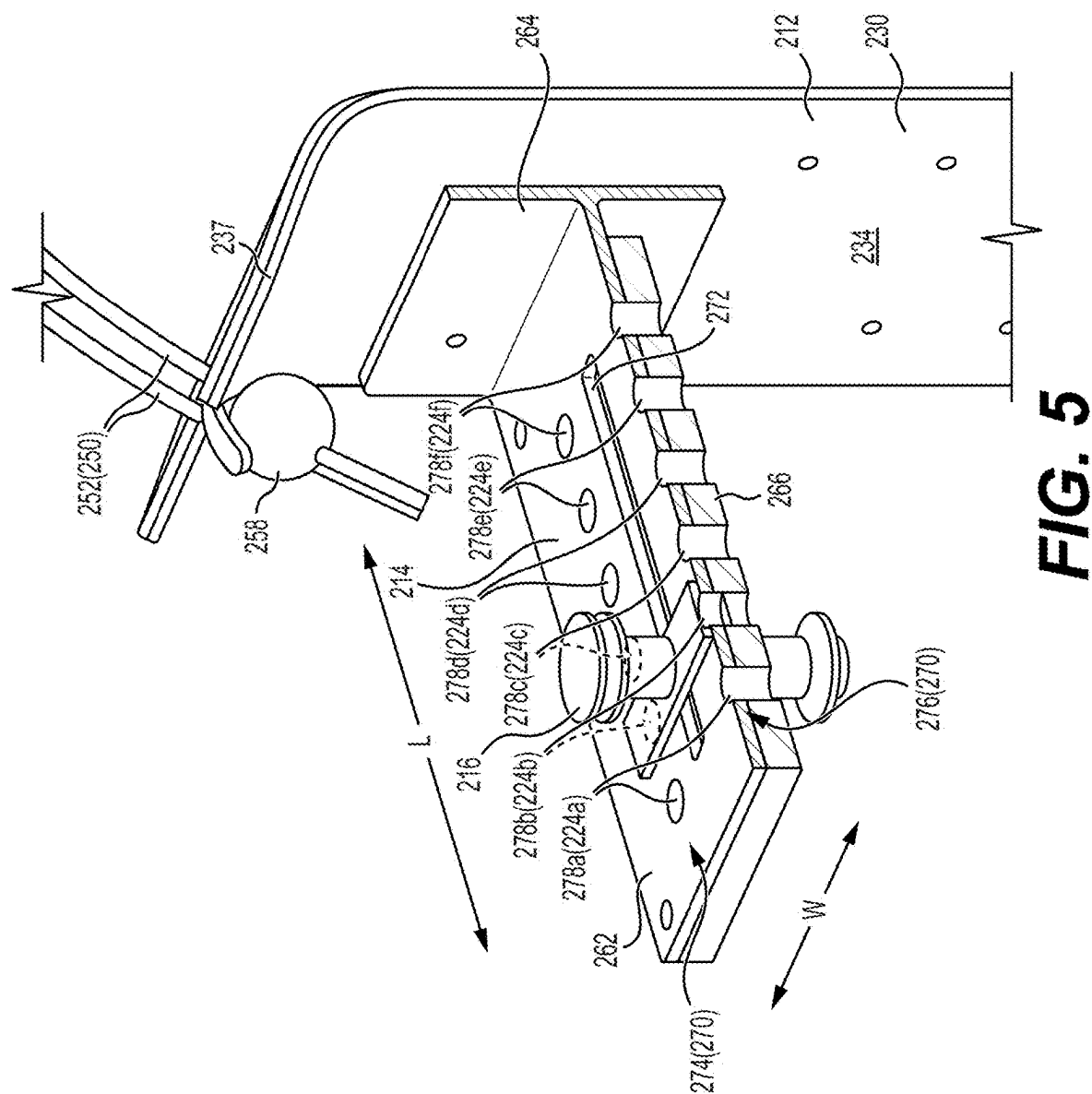
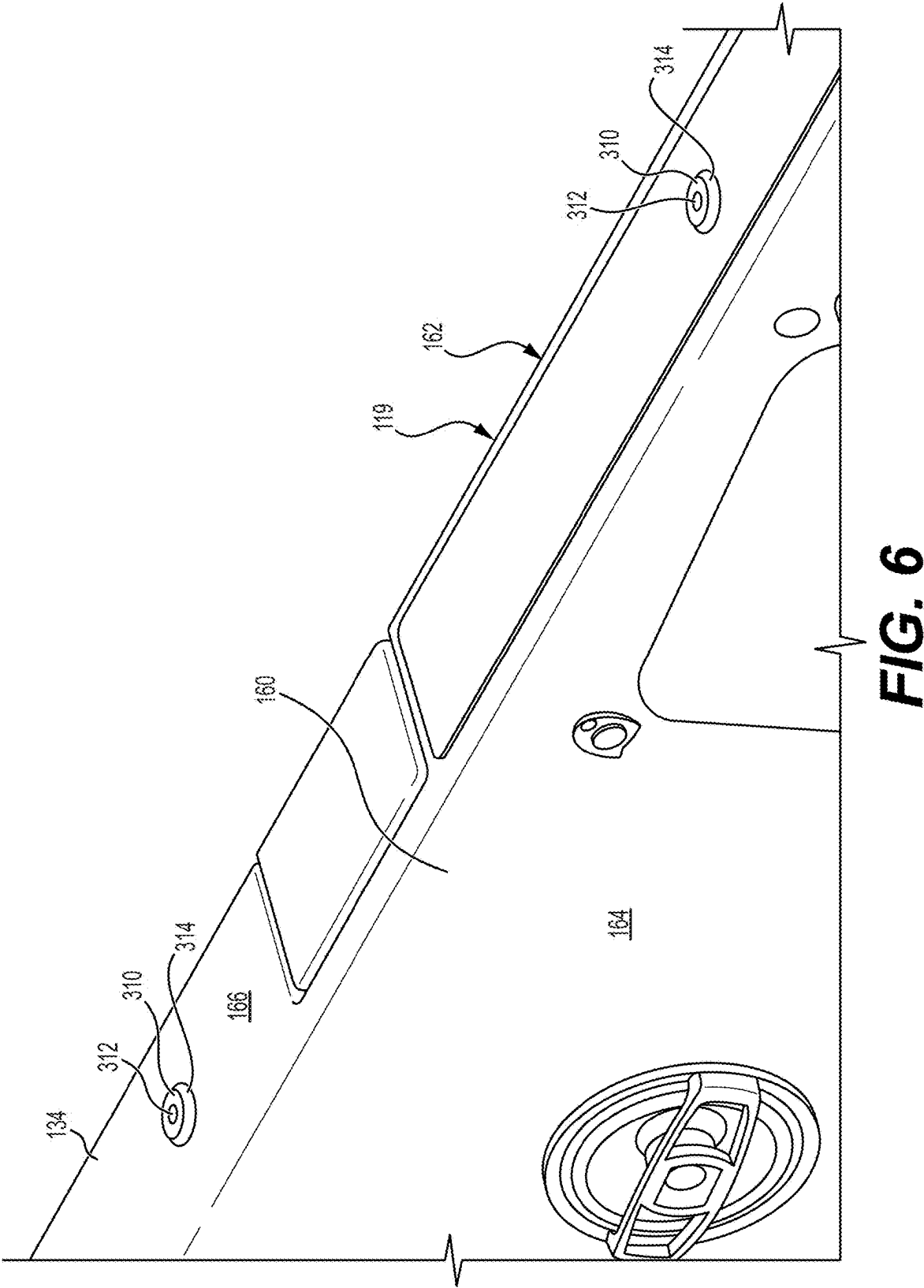


FIG. 4B





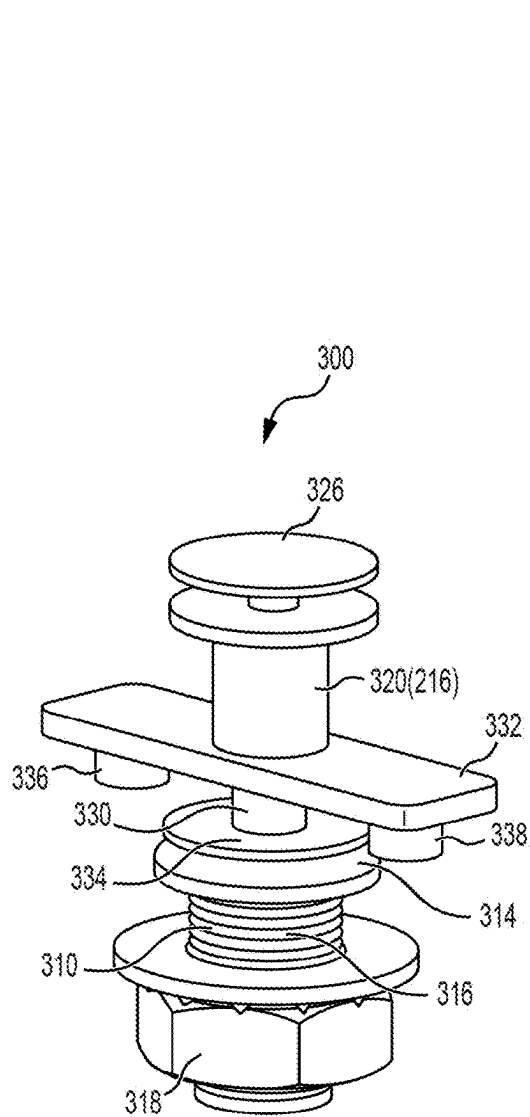


FIG. 7A

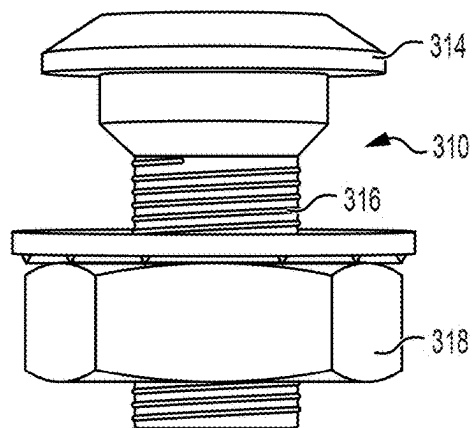
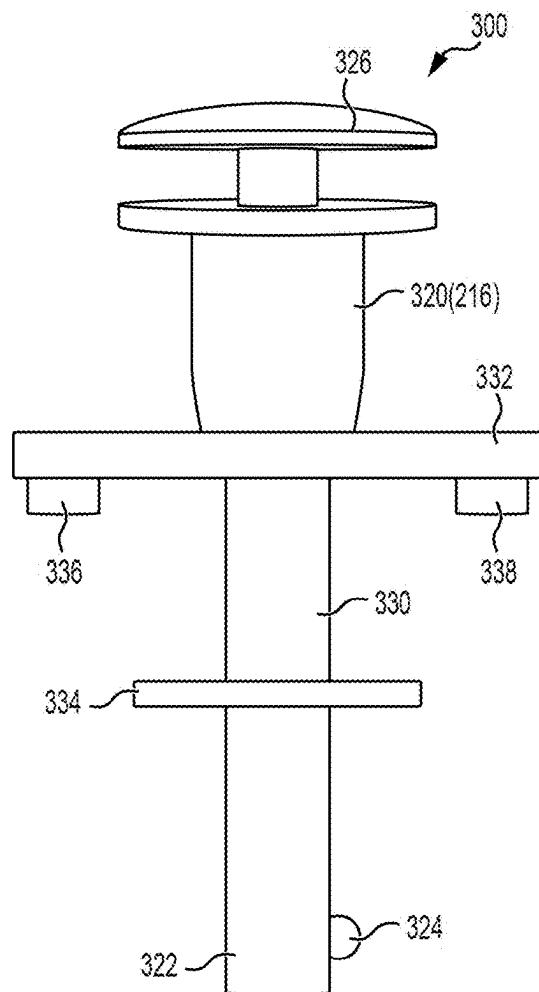


FIG. 7B

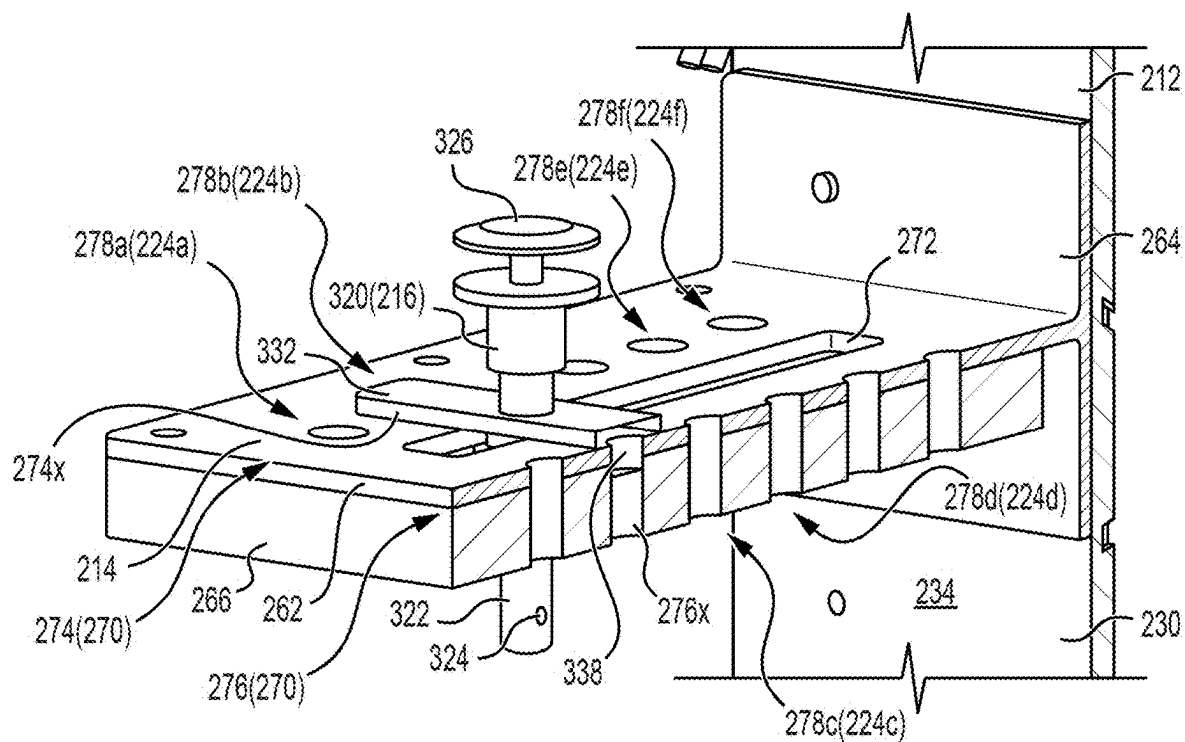


FIG. 8A

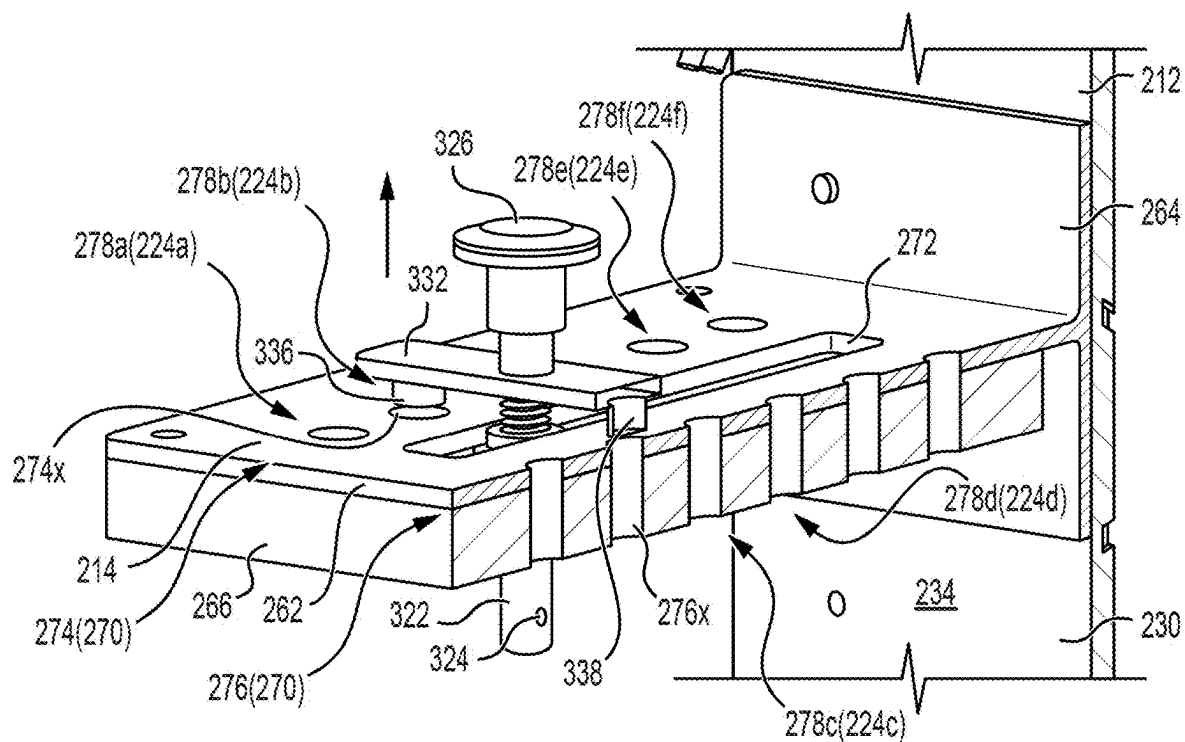


FIG. 8B

FIG. 8D

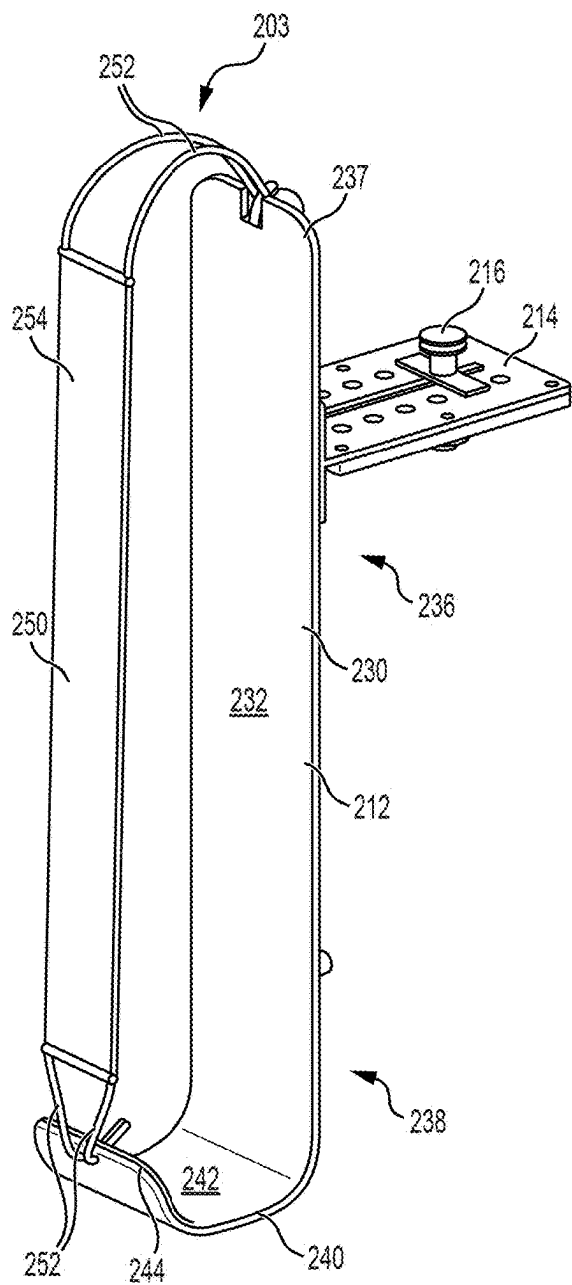


FIG. 9A

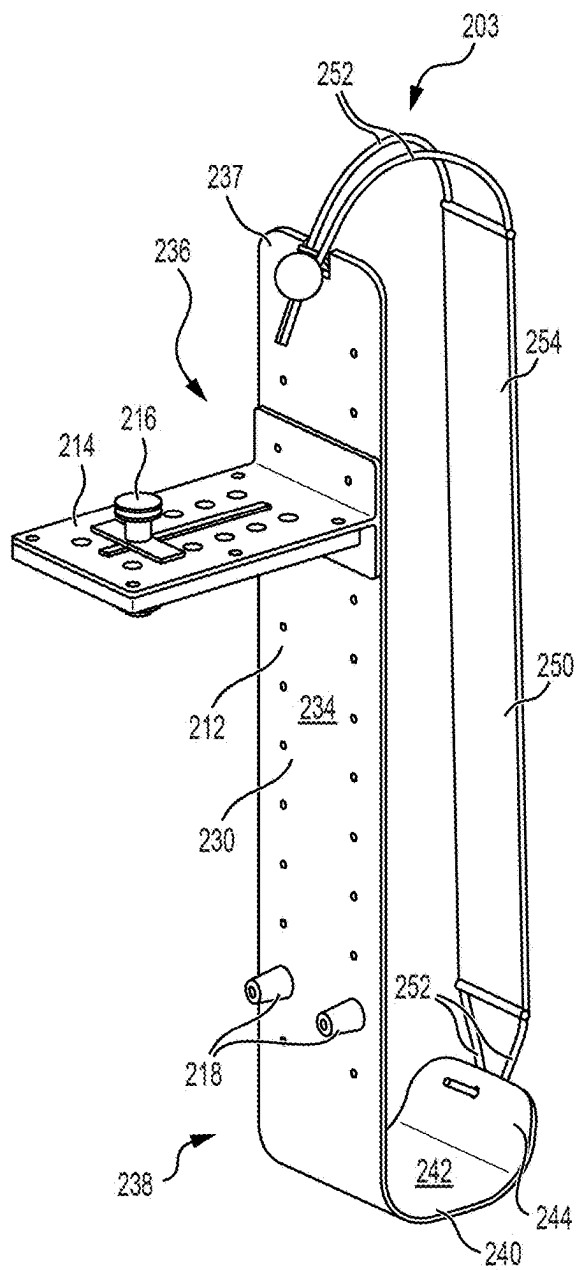


FIG. 9B

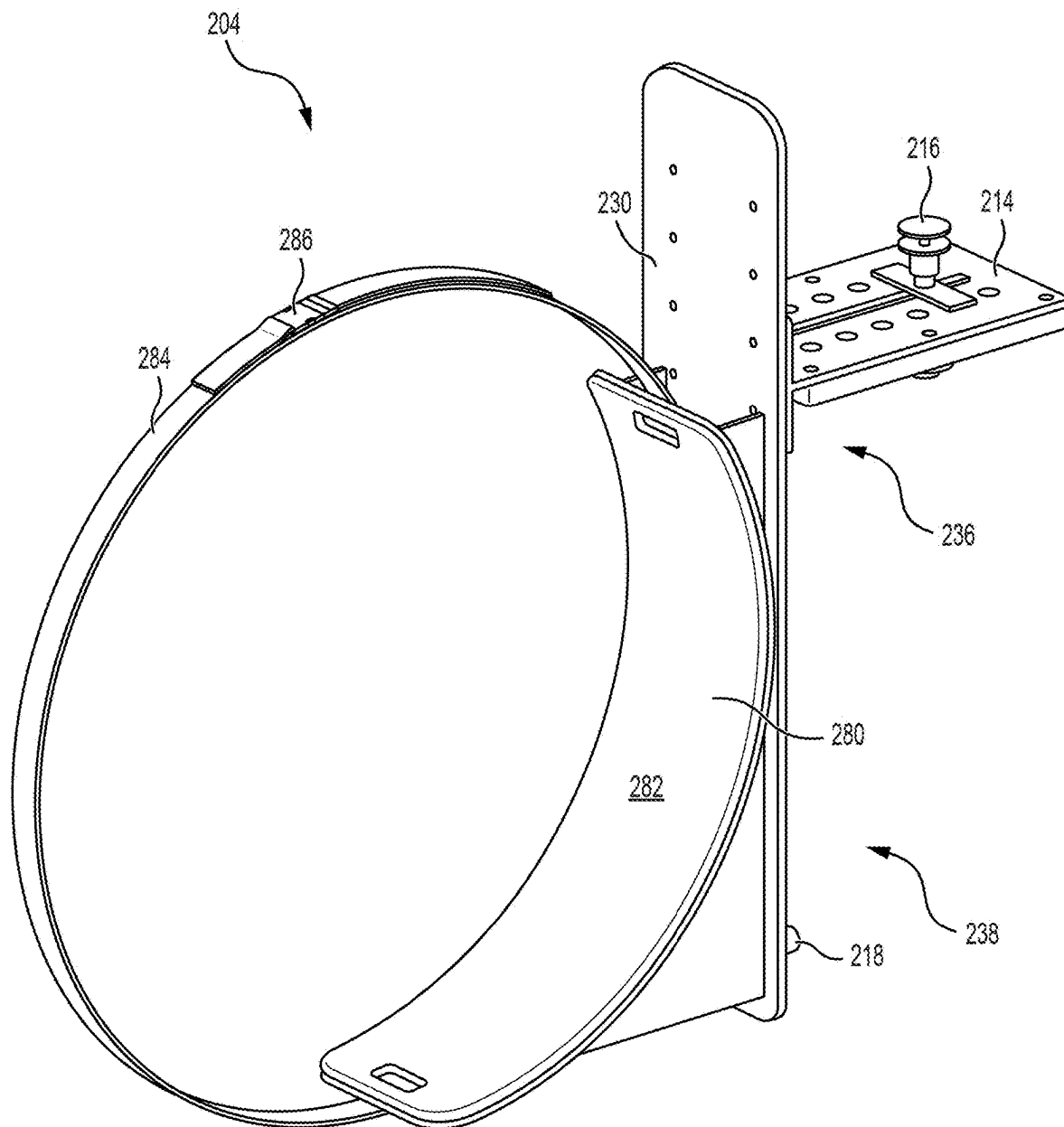


FIG. 10A

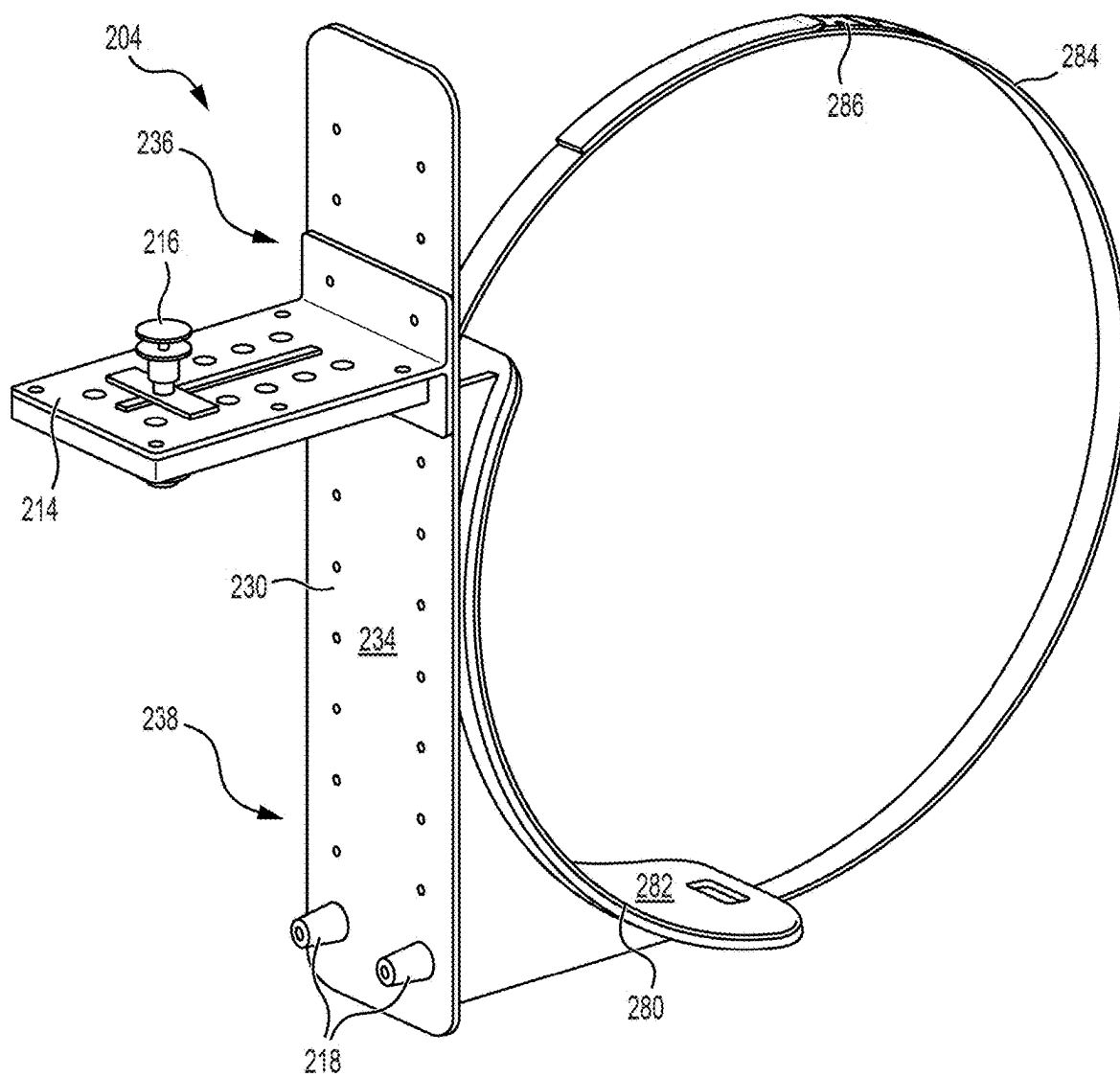


FIG. 10B

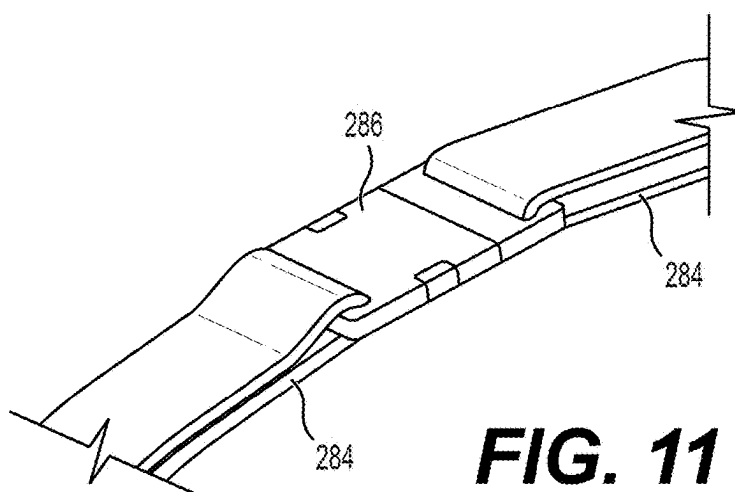


FIG. 11

ADJUSTABLE EQUIPMENT STORAGE RACK FOR A BOAT

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application No. 63/562, 897, filed Mar. 8, 2024, and titled “WATER-SPORT EQUIPMENT STORAGE RACK FOR A BOAT,” the entirety of which is incorporated herein by reference.

FIELD OF THE INVENTION

[0002] The present disclosure generally relates to an apparatus for storing equipment on a boat, particularly, water-sport equipment on recreational boats used for watersports.

BACKGROUND OF THE INVENTION

[0003] Watersport boats often are used to tow watersport participants, such as water skiers, wakeboarders, and the like, using a tow-line. For water skiing and wakeboarding, the participant holds onto one end of the tow-line and the other end is attached to the boat. For tubing, the tow-line is attached to the tube and the watersport participant(s) holds onto the tube. A boat may also be used to generate a wake on which a watersport participant, such as a wake surfer or foiler, may wake surf or foil, generally without holding onto a tow-line once they get going. Another water activity that may be enjoyed with recreational boats is paddleboarding, such as stand-up paddleboarding, which involves standing up on a relatively large board (i.e., a paddleboard) and using a paddle to make way through the water.

[0004] Buoyant water mats, which are also known as water pads, lily pads, or floating islands, are mats that float on the surface of a body of water, such as a pond, lake, or river, and can support one or more persons. Water mats are typically large, rectangular mats that may have a width of 6 feet and a length longer than 6 feet, such as, for example, 12 feet, 15 feet, or 18 feet. With such a large, buoyant area, these mats have a capacity to hold a substantial amount of weight and can suitably hold multiple people at once, such as, for example, four, six, or eight adults. These water mats may be rolled up from their flat, rectangular deployed position in order to be stowed.

SUMMARY OF THE INVENTION

[0005] In one aspect, the invention relates to an equipment storage rack for a boat. The equipment storage rack can be attached to a barrier and can be adjustable relative to the barrier.

[0006] In another aspect, the invention relates to an equipment storage rack for a boat that can store a piece of equipment on the exterior of the boat. The equipment storage rack can be attached to an upper perimeter surface of a barrier of the boat and adjustable for different widths of the upper perimeter surface.

[0007] In a further aspect, the invention relates to an adjustable equipment storage rack for a boat. The adjustable equipment storage rack includes a cradle for supporting a piece of equipment, and a mounting bracket connected to the cradle to support the cradle. The mounting bracket has a plurality of engagement positions for a fastener to engage with the mounting bracket and to secure the mounting bracket to the boat at a fastening position. The cradle is a

different distance from the fastening position at each engagement position of the plurality of engagement positions.

[0008] In still another aspect, the invention relates to a boat including the equipment storage rack of any of the preceding aspects.

[0009] In yet another aspect, the invention relates to a boat including a barrier and an adjustable equipment storage rack mounted to the barrier. The adjustable equipment storage rack includes a cradle assembly. The cradle assembly includes a cradle for supporting a piece of equipment and a mounting bracket connecting the cradle to the barrier. The mounting bracket is positionable in a plurality of cradle positions and, at each position of the plurality of cradle positions, the cradle is a different distance from the barrier.

[0010] These and other aspects of the invention will become apparent from the following disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 shows a boat equipped with an adjustable equipment storage rack.

[0012] FIG. 2 shows a portion of the starboard side of the hull of the boat of FIG. 1 with the adjustable equipment storage rack connected thereto.

[0013] FIGS. 3A and 3B are stern views of a barrier with the adjustable equipment storage rack attached thereto. FIG. 3A shows a fastener engaged with a mounting bracket at a second engagement position and FIG. 3B shows the fastener engaged with the mounting bracket at a fourth engagement position.

[0014] FIGS. 4A and 4B are perspective views of a cradle assembly. FIG. 4A shows a front side of the cradle assembly and FIG. 4B shows a back side of the cradle assembly.

[0015] FIG. 5 is a detail view of the mounting bracket of the cradle assembly shown in FIGS. 4A and 4B.

[0016] FIG. 6 is a detail view of a barrier of the boat with the cradle assemblies of the adjustable equipment storage rack removed.

[0017] FIGS. 7A and 7B show a fastener assembly that can be used with the adjustable equipment storage rack. FIG. 7A shows a quick-release fastener connected to a receiver and FIG. 7B shows the quick-release fastener removed from the receiver.

[0018] FIGS. 8A to 8D illustrate the process of adjusting the position of the cradle assembly from the cradle position shown in FIG. 3A to the cradle position shown in FIG. 3B. FIG. 8A shows the mounting bracket and fastener engaged to position the cradle assembly in the cradle position shown in FIG. 3A. FIG. 8B shows a step of disengaging the fastener. FIG. 8C shows a step of moving the mounting bracket. FIG. 8D shows a step of engaging the fastener with the mounting bracket to position the cradle assembly in the cradle position shown in FIG. 3B.

[0019] FIGS. 9A and 9B are perspective views of another cradle assembly that can be used with the adjustable equipment storage rack. FIG. 9A shows a front side of the cradle assembly and FIG. 9B shows a back side of the cradle assembly.

[0020] FIGS. 10A and 10B are perspective views of another cradle assembly that can be used with the adjustable equipment storage rack. FIG. 10A shows a front side of the cradle assembly and FIG. 10B shows a back side of the cradle assembly.

[0021] FIG. 11 is a detail view of a clasp of a strap for the cradle assembly shown in FIGS. 10A and 10B.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0022] As noted above, recreational boats may be used for watersports and other water-based recreational activities. These water activities use various equipment, such as boards (for example, paddleboards, wakeboards, and surfboards), water skis, or water mats, which are referred to generically herein as water-sport equipment. When not in use, this watersport equipment is often stored on the boat, preferably in racks or compartments. Preferably, these racks and compartments are designed to maximize the space onboard the boat and prevent the equipment from moving around while stored.

[0023] The present disclosure generally relates to recreational boats and an apparatus that can be connected to or attached to the boat for storing watersport equipment. More specifically, the apparatus for storing watersport equipment is a rack that can be positioned on the boat to store the watersport equipment on an exterior of the boat, maximizing the space within the interior of the boat. In addition, the rack can be an adjustable rack capable of being attached to surfaces and features, such as barriers on the boat, of different widths.

[0024] FIG. 1 shows a boat 100 having an adjustable equipment storage rack 200. FIG. 1 is a perspective view of the boat 100, showing the stern of the boat 100. The boat 100 of this embodiment includes a hull 110 having a bow 111, a transom 113, a hull bottom 115, a port-side hull 117, and a starboard-side hull 119. The hull bottom 115 includes the portions of the hull 110 between the chines. The boat 100 has a longitudinal centerline 102 running down the middle of the boat 100, halfway between the port-side hull 117 and the starboard-side hull 119. The boat 100 shown in FIG. 1 is a bowrider having both a bow seating area 122 positioned in the bow 111 of the boat 100 and a primary seating area 124 (sometimes also referred to as the cockpit) positioned aft of a windshield (not shown). Within the boat's interior 120 is a control console 152 for operating the boat 100. Here, the control console 152 is positioned on the starboard side of the boat 100, proximate to and aft of the windshield. Although described in reference to a bowrider, the embodiments discussed herein may be used with any suitable boat including cuddies, center consoles, and cruisers, for example. Various embodiments discussed herein may also be suitable for use with other boats such as pontoon boats.

[0025] Collectively, the bow 111, the transom 113, the port-side hull 117, and starboard-side hull 119 define an interior 120 of the boat 100 with an exterior 130 of the hull 110 being the space outward from the hull 110. The port-side hull 117 and the starboard-side hull 119 have a port gunwale and a starboard gunwale, respectively, and in some embodiments, the boat can include a port coaming and a starboard coaming formed on the port gunwale and the starboard gunwale, respectively. The hull side (e.g., the port-side hull 117 or the starboard-side hull 119) and, when present, the coaming, form a perimeter 132 of the boat 100 with a top surface 134 that is generally horizontal. As depicted, this top surface is a generally horizontal deck surface, but this surface can also be formed as part of the hull. This surface is referred to herein as a top perimeter surface of the deck or hull.

[0026] The boat 100 can include a swim platform 106 connected to the stern to make it easier for people to get into the water from the boat 100 or into the boat 100 from the water. The swim platform 106 can be horizontal and should be capable of supporting a human and preferably is capable of supporting at least 500 lbs., and, even more preferably, 1250 lbs. The swim platform 106 can be constructed from any suitable material that can be used in a marine environment including, for example, fiberglass or teak. The swim platform 106 can be integrally formed with the stern of the boat 100 (as shown in FIG. 1), but the swim platform 106 also can be an attachable/detachable platform that is attached to the transom 113 of the boat 100 using, for example, two brackets screwed to the transom 113.

[0027] The boat 100 is equipped with an apparatus for towing a watersports participant. As shown in FIG. 1, the towing apparatus is integrated into a hardtop cover 142, but other suitable towing apparatuses can be used, including, for example, the towers shown and described in U.S. Pat. Nos. 9,580,155 and 10,150,540, which are incorporated by reference herein in their entireties. The hardtop cover 142 is supported by two legs: a port leg 144 and a starboard leg 146. The port leg 144 is attached to the boat 100 on the port side of the longitudinal centerline 102 of the boat 100 and the starboard leg 146 is attached to the boat 100 on the starboard side of the longitudinal centerline 102 of the boat 100. Preferably, each of the port leg 144 and the starboard leg 146 is attached to an upper surface (such as gunwales) above the port-side hull 117 and the starboard-side hull 119, respectively. The hardtop cover 142 is connected to an upper portion of each of the port leg 144 and the starboard leg 146 and spans the interior 120 of the boat 100 at a height suitable for passengers to pass underneath while standing. The hardtop cover 142 also extends forward and aft over a portion of the primary seating area 124 to provide shade. In addition, the hardtop cover 142 has a tow-line-attachment structure 148 at an aft portion of the hardtop cover 142. This tow-line-attachment structure 148 can be used to connect a tow-line suitable for towing a watersports participant, such as a wake surfer. Any suitable tow-line-attachment structure can be used, including, but not limited to, the integrated light and tow-line-attachment assembly disclosed in U.S. Pat. No. 6,539,886, which is incorporated by reference herein in its entirety.

[0028] The boat 100 can include other features for watersports. For example, although not depicted herein, the boat 100 can include the capability to add ballast. For example, the ballast systems shown and described in U.S. Patent Application Pub. No. 2024/0326957, which is incorporated by reference herein in its entirety, can be used in the boat 100. The boat 100 can include other features and configurations for watersports, such as the operation of the propulsion system described in the '957 publication, or other surf devices, such as, for example, the port and starboard wake-modifying devices disclosed in U.S. Patent No. 8,833,286, which is incorporated by reference herein in its entirety. Other examples of suitable alternative surf devices are shown and described in U.S. Pat. Nos. 9,802,684 and 10,358,189 and the surf devices described in U.S. Patent Application Publication 2022/0089258, the disclosures of which are incorporated by reference herein in their entirety.

[0029] The boat 100 includes a propulsion system, including one or more propulsion units 150. As depicted in FIG. 1, the boat 100 can include a plurality of propulsion units 150.

The propulsion units **150** depicted in FIG. 1 are each outboards, but other types of propulsion systems can be used, including, for example, inboards, sterndrives, and jet drives. The propulsion unit **150** includes a motor **154** operatively coupled to a propulsor to drive the propulsor. In this embodiment, the motor **154** is a combustion engine, but other suitable motors can be used, including electric motors. The propulsor of this embodiment is a propeller **156**, but other suitable propulsors can be used, such as, for example, impellers in jet drives. The motor **154** is configured to drive (rotate) the propeller **156** through one or more drive shafts coupling the motor **154** to the propeller **156**. Rotating the propeller **156** drives the boat **100** through the water. To turn the boat **100**, an operator can turn a steering wheel at the control console **152** to rotate and angle the outboard motor (both the motor **154** and the propeller **156**) relative to the longitudinal centerline **102**.

[0030] As noted above, the boat **100** is equipped with an adjustable equipment storage rack **200**. The adjustable equipment storage rack **200** includes a cradle **212** (FIG. 2) that is shaped and sized to hold a piece of equipment **10**, such as a piece of the watersport equipment. The piece of equipment **10** can be a board (for example, a paddleboard, a wakeboard, or a surfboard), a water ski, or a water mat. The piece of equipment **10** shown in the accompanying figures is a paddleboard. The adjustable equipment storage rack **200** can include one or more cradles **212**. A plurality of cradles **212**, such as two cradles **212**, can be used to support relatively large or long pieces of watersport equipment, like paddleboards. To maximize the space within the interior **120**, the cradles **212** of the adjustable equipment storage rack **200** are positioned on the exterior **130** of the hull **110**. Positioning the cradles **212** and the piece of equipment **10** stored therein with a majority of each cradle **212** and/or the piece of equipment **10** lower than the upper surface of the side of the deck or hull, such as lower than the top surface **134** of the perimeter **132**, has certain advantages to visibility and the like. Such a positioning provides better visibility for the driver to watch for other boaters or people in the water, as compared to other racks that have the piece of equipment stored above the top surface **134**. These other racks have a “truck topper” that makes it so the driver or others in the boat cannot see port aft or starboard aft past the port leg **144** or the starboard leg **146** in most situations. The piece of equipment effectively forms a wall, blocking visibility.

[0031] FIG. 2 shows a portion of the starboard-side hull **119** with the adjustable equipment storage rack **200** connected thereto. The adjustable equipment storage rack **200** can include one or more cradle assemblies. The adjustable equipment storage rack **200**, depicted in FIG. 2, includes a plurality of cradle assemblies, including a first cradle assembly **201** and a second cradle assembly **202**. The plurality of cradle assemblies can be used to better support a large piece of equipment **10**, such as a paddle board or other boards. The following discussion will refer to the first cradle assembly **201**, but the discussion applies equally to other cradle assemblies of the plurality of cradle assemblies, such as the second cradle assembly **202**.

[0032] The boat **100** includes a barrier **160** and the adjustable equipment storage rack **200** is mounted to the barrier **160**. The barrier **160** can be various barriers on the boat **100**, such as a bulkhead or other structure separating compartments within the boat **100**. The barrier **160** can be, for example, a barrier **160** that defines the interior **120** of the

boat **100** and separates the interior **120** from the exterior **130** of the boat **100**. On a pontoon boat, for example, the barrier can be a fence. In the boat **100** depicted in FIG. 2, for example, the barrier **160** includes a portion of the hull such as one of the port-side hull **117** (FIG. 1) or the starboard-side hull **119**. In FIG. 1, the adjustable equipment storage rack **200** is shown as being mounted to the port-side hull **117**, but in FIG. 2, the adjustable equipment storage rack **200** is mounted to the starboard-side hull **119**. The barrier **160** includes an outward facing surface, which, as depicted in FIG. 2, is an exterior surface **162** of the barrier **160**. The barrier **160** also includes a surface opposite of the outward facing surface, such as an inward facing surface, which, as depicted in FIG. 2, is an interior surface **164**. The barrier **160** also includes a top surface **166**. The top surface **166** can be the top surface **134** of the perimeter **132** of the boat **100** as discussed above.

[0033] The first cradle assembly **201** includes a cradle **212** that is sized and shaped to support the piece of equipment **10**, such as by directly supporting the piece of equipment **10**. The cradle **212** can have various shapes, such as a J-shape (depicted), concave shapes, circular or arcuate shapes, U-shapes, V-shapes, or L-shapes. The cradle **212** can be sized and shaped depending upon the piece of watersport equipment to be stored therein. For example, the rack can be smaller for wakeboards or surfboards or larger for paddle boards. In another example, the rack can have a larger horizontal area when storing a rolled-up water mat or different shape. Likewise, although shown in the accompanying figures as storing only one piece of watersport equipment, the rack can have shapes that allow multiple pieces of watersport equipment to be stored.

[0034] The first cradle assembly **201** also includes a mounting bracket **214** connected to the cradle **212** to support the cradle **212**. The mounting bracket **214** connects the cradle **212** to the barrier **160**. The mounting bracket **214** can be attached to the barrier **160** at different locations. For example, the mounting bracket **214** can be attached to the top surface **166** of the barrier **160**. More specifically, the mounting bracket **214** can be directly attached to the top surface **166** of the barrier **160**. The first cradle assembly **201** can include a fastener **216**. The fastener **216** can engage with the mounting bracket **214** and secure the mounting bracket **214** to the boat **100**. The fastener **216** can secure the mounting bracket **214** to the barrier **160**, such as the top surface **166** of the barrier **160**.

[0035] FIGS. 3A and 3B are stern views of the barrier **160** and the adjustable equipment storage rack **200** attached thereto. The fastener **216** engages with the fastener **216** and secures the mounting bracket **214** to the barrier **160**, such as the top surface **166**, at a fastening position **222**. As noted above, the adjustable equipment storage rack **200** is adjustable. The adjustable equipment storage rack **200** can be positioned in a plurality of different positions. More specifically, the fastener **216** can engage with the mounting bracket **214** at a plurality of engagement positions **224**. At each engagement position **224** of the plurality of engagement positions **224**, the cradle **212** is a different distance from the fastening position **222**. For example, FIG. 3A shows the fastener **216** engaged with the mounting bracket **214** at a second engagement position **224b**, and FIG. 3B shows the fastener **216** engaged with the mounting bracket **214** at a fourth engagement position **224d**. The cradle **212** is farther from the fastening position **222** when the fastener

216 is engaged with the mounting bracket 214 at the second engagement position 224b than when the fastener 216 is engaged with the mounting bracket 214 at the fourth engagement position 224d.

[0036] The cradle 212 is preferably positioned to abut the barrier 160, such as the exterior surface 162. Such a positioning prevents vibration of the piece of equipment 10 in the cradle 212. By having a plurality of engagement positions 224, the cradle 212 can abut the exterior surface 162 evenly and accommodate barriers with different widths and different distances from the fastening position 222 to the exterior surface 162.

[0037] FIGS. 4A and 4B are perspective views of the first cradle assembly 201. FIG. 4A shows a front side of the first cradle assembly 201 and FIG. 4B shows a back side of the first cradle assembly 201. As noted above, the first cradle assembly 201 includes a cradle 212. The cradle 212 can have various shapes, such as the J-shape shown in FIGS. 4A and 4B. The cradle 212 includes a first leg and a second leg oriented transversely to the first leg. The first leg can be an upright leg 230 and the second leg can be a lateral leg 240. The upright leg 230 includes an equipment-facing surface 232 and the lateral leg 240 also includes an equipment-facing surface 242. The equipment-facing surfaces 232, 242 can directly support the piece of equipment 10 (FIGS. 3A and 3B), such that the piece of equipment 10 abuts the equipment-facing surfaces 232, 242 when placed in the cradle 212. When these equipment-facing surfaces 232, 242 directly support the piece of equipment 10, padding, such as foam padding, can be placed on these surfaces to protect the piece of equipment 10 from damage.

[0038] The upright leg 230 also includes a back side surface 234. The back side surface 234 is a surface opposite the equipment-facing surface 232 and faces away from the piece of equipment 10. (FIGS. 3A and 3B) The back side surface 234 can face toward the barrier 160, such as toward the exterior surface 162. The upright leg 230 can be oriented in a generally vertical orientation when attached to the barrier 160, and the lateral leg 240 can be oriented in a horizontal vertical orientation when attached to the barrier 160. The upright leg 230 has an upper portion 236 and a lower portion 238. The upper portion 236 can be an upper half, an upper third, or an upper quarter of the upright leg 230. Similarly, the lower portion 238 can be a lower half, a lower third, or a lower quarter of the upright leg 230. The lateral leg 240 is attached, such as directly attached, to the lower portion 238 of the upright leg 230. More specifically, the lateral leg 240 is attached to a lower end of the upright leg 230. The upright leg 230 and the lateral leg 240 can be a continuous piece, such as a continuous plate, and the lateral leg 240 can be formed by being bent relative to the upright leg 230.

[0039] With the upright leg 230 and the lateral leg 240 positioned as discussed above and as shown in FIGS. 3A and 3B, the upright leg 230 can support a broad surface of the piece of equipment 10 and the lateral leg 240 can support a narrower surface of the piece of equipment 10, when the piece of equipment 10 is located in the cradle 212. As shown in FIG. 4A and 4B, the cradle 212 has a J-shape. The lateral leg 240 has a distal end 244, which is an end of the lateral leg 240 distal from the upright leg 230. The distal end 244 is bent or otherwise curved toward the piece of equipment 10 (FIGS. 3A and 3B). When mounted to the barrier 160, the distal end 244 can be curved upward in the direction that the

upright leg 230 extends from the lateral leg 240. This upward bend of the distal end 244 helps retain and secure the bottom of the piece of equipment 10 in the cradle 212.

[0040] The upright leg 230 includes an upper end 237. The upper end 237 of the upright leg 230 can be curved or bent. The upper end 237 can be curved or bent in the direction that the back side surface 234 faces. As will be discussed further below, the mounting bracket 214 is attached to the back side surface 234 and extends from the back side surface 234. The upper end 237 can be bent in the direction that the mounting bracket 214 extends. The bent portion of the upper end 237 can have, for example, an angle with the back side surface 234 that is from 110 degrees to 150 degrees. As shown in FIGS. 3A and 3B, for example, the upper end 237 can be bent toward the barrier 160. When the cradle 212 is located on the exterior 130 of the boat 100, the piece of equipment 10 can be placed into the cradle 212 by a person located in the interior 120 of the boat 100. Bending the upper end 237 towards the interior 120 helps the person insert or remove the piece of equipment 10 from the cradle 212.

[0041] To help protect the exterior surface 162 of the barrier 160, such as the starboard-side hull 119, one or more bumpers 218 can be attached, such as directly attached, to the back side surface 234 of the upright leg 230. As shown in FIG. 4B, the bumpers 218 can be located on the lower portion 238 of the upright leg 230. The bumpers 218 can be elastomeric bumpers, such as rubber bumpers attached by a suitable means to the lower portion 238, such as by a fastener, like a screw, or adhesive.

[0042] As illustrated in FIGS. 3A to 4B, the piece of equipment 10 can be further secured into the cradle 212 by one or more straps 250. Each strap 250 can be engageable with the cradle to secure the piece of equipment 10 to the cradle 212. The strap 250 can have one end attached to the distal end 244 of the lateral leg 240 and another end attached to the upper end 237 of the upright leg 230. The strap 250 can comprise, at least in part, an elastomeric material, such as bungee cord 252. In the depicted embodiment, the strap 250 also includes a sheath 254, such as a canvas sheath. The bungee cord 252 attaches to or extends through the sheath 254. With the ends of the strap 250 attached as discussed above, the strap 250 compresses the piece of equipment 10 against the cradle 212 and secures the piece of equipment 10 in the cradle 212.

[0043] The strap 250 can be detachably connected to the cradle 212, such as detachably connected to the upper end 237 of the upright leg 230. The upper end 237 of the upright leg 230 can include a slot 256, and a portion of the strap 250, such as a portion of the bungee cord 252, extends through the slot 256. A cord stop 258 can be connected to the bungee cord 252. The cord stop 258 can be a cord lock, such as a cord lock with a spring-loaded clasp that grips the cord. The cord stop 258 can be sized to have at least one dimension larger than the slot 256 to prevent the cord stop 258 from moving through the slot 256, thus securing the bungee cord 252 in the slot 256. The elastic force of the strap 250 pulls the cord stop 258 against the back side surface 234 to hold the cord stop 258 in place. When a cord lock is used, the length of the strap can be adjusted to secure pieces of equipment 10 (e.g., boards) of different sizes and apply the elastic force to the piece of equipment 10 and the cord stop 258. To release the strap 250, the user can pull on the end of the strap 250 with the cord stop 258 to move the cord stop

258 away from the back side surface 234 and then slide the bungee cord 252 out of the slot 256.

[0044] As shown in FIGS. 4A and 4B, the mounting bracket 214 is connected to the cradle 212, such as attached to the cradle 212. More specifically, the mounting bracket 214 is attached to the back side surface 234, such as directly attached to the back side surface 234 of the upright leg 230. The mounting bracket 214 is attached to the upper portion 236 of the upright leg 230 to position the cradle 212 with a majority of the cradle 212 lower than the top surface 166 of the exterior surface 162 (see FIGS. 3A and 3B). The mounting bracket 214 can be formed as a plate (i.e., a mounting bracket plate 262) with a mounting bracket flange 264 formed on one end of the mounting bracket plate 262. The mounting bracket flange 264 can be directly attached to the back side surface 234 of the upright leg 230 using, for example, fasteners. The mounting bracket flange 264 can be attached to the back side surface 234 at a plurality of different heights to accommodate different heights of the barrier 160. For example, the cradle 212 is preferably positioned high enough that the lateral leg 240 is out of the water when positioned on the exterior 130 of the boat 100.

[0045] FIG. 5 is a detail view of the mounting bracket 214. To help protect the barrier 160, such as the top surface 166 of the barrier 160, a pad 266 can be located on an underside of the mounting bracket plate 262. The pad 266 can be, for example, a plastic pad, such as high-density polyethylene.

[0046] The mounting bracket 214 shown in FIG. 5 includes a slot 272. The slot 272 is an elongated slot that is elongated in a longitudinal direction L of the slot 272. The slot 272 also includes a width direction W, which is a direction that is perpendicular to the longitudinal direction L of the slot 272. The mounting bracket 214 is also oriented in the longitudinal direction L and the width direction W. The fastener 216 can engage with the mounting bracket 214 by being located in the slot 272. For example, the slot 272 extends through the mounting bracket plate 262 and the pad 266 and the fastener 216 can extend through the slot 272 to engage with a receiver 310 (FIG. 6) located in the barrier 160 (FIG. 6), thus connecting the mounting bracket 214 to the barrier 160. As will be discussed further below, the fastener 216 can slide within the slot when the mounting bracket 214 is moved between the plurality of engagement positions 224.

[0047] The mounting bracket 214 includes a plurality of openings 270. Each opening of the plurality of openings 270 can be a hole, such as a blind hole or a through hole. Some of the plurality of openings 270 are in a first set of openings 274, and some of the plurality of openings 270 are in a second set of openings 276. The openings of the first set of openings 274 are arrayed in a linear array on one side of the slot 272, and the openings of the second set of openings 276 are arrayed in a linear array on an opposite side of the slot 272 in the width direction W. The openings of the first set of openings 274 and the openings of the second set of openings 276 are arranged to form corresponding pairs of openings 278. Each pair of openings 278 includes a first opening 274x from the first set of openings 274 and a second opening 276x from the second set of openings 276. The first opening 274x is thus located on one side of the slot 272 and the second opening 276x is located on an opposite side of the slot 272 in the width direction W. Each pair of openings 278 corresponds to one engagement position of the plurality of engagement positions 224. Similarly, each opening of the

first set of openings 274 corresponds to one engagement position of the plurality of engagement positions 224, and each opening of the second set of openings 276 corresponds to another engagement position of the plurality of engagement positions 224. For example, as depicted in FIG. 5, the mounting bracket 214 includes six engagement positions and corresponding pairs of openings 278. These different positions, positions one to six, are shown in the figures with the reference character a to f appended to reference numerals 224 and 274 to indicate the first through sixth position, respectively.

[0048] FIG. 6 is a detail view of the barrier 160 with the first cradle assembly 201 and the second cradle assembly 202 is removed. The barrier 160 includes one or more receivers 310 located in the barrier 160. In FIG. 6, the barrier 160 includes two receivers, one for the first cradle assembly 201 (FIG. 2) and one for the second cradle assembly 202 (FIG. 2). The receiver 310 can be embedded in the barrier 160, such as in the top surface 166 of the barrier 160. The receiver 310 includes a receiver bore 312 to receive the fastener 216. The receiver 310 can also include a receiver flange 314 to assist in attaching the receiver 310 to the top surface 166 of the barrier 160.

[0049] FIGS. 7A and 7B show a fastener assembly 300 that can be used with the adjustable equipment storage rack 200. The fastener assembly 300 includes the receiver 310 and a quick-release fastener 320 that can be used as the fastener 216 discussed herein. Although described as a quick-release fastener 320, the fastener 216 can be another type of fastener such as other bolts, pins, and the like. FIG. 7A shows the quick-release fastener 320 connected to the receiver 310 and FIG. 7B shows the quick-release fastener 320 removed from the receiver 310.

[0050] The receiver 310 shown in FIGS. 7A and 7B includes a receiver shaft 316 extending from the receiver flange 314. The receiver bore 312 (FIG. 6) is formed in the receiver shaft 316. The receiver shaft 316 can include threads formed on an exterior thereof, and a receiver nut 318 can be threaded onto the receiver shaft 316 to secure the portion of the barrier 160 with the top surface 166 between the receiver nut 318 and the receiver flange 314.

[0051] The quick-release fastener 320 includes a fastener shaft 322. The fastener shaft 322 engages with the receiver 310. More specifically, the fastener shaft 322 is inserted into the receiver bore 312. The quick-release fastener 320 includes a catch 324 movable between a latched position and an unlatched position. The latched position is depicted in FIG. 7B with the catch 324 extending from an outer surface of the fastener shaft 322. In the unlatched position, the catch 324 can be retracted into the fastener shaft 322. The catch 324 of the quick-release fastener 320 can be a detent, for example. When in the latched position, the catch 324 engages with, for example, a recess or a slot (not shown) formed on an inner surface of the receiver bore 312 to prevent the fastener shaft 322 from being withdrawn from the receiver bore 312. To withdraw the fastener shaft 322 from the receiver bore 312, the catch 324 is moved to the unlatched position. A biasing member, such as a spring, can be used to exert a biasing force on the catch 324 to push the catch 324 toward the latched position and hold the catch 324 in the latched position.

[0052] The quick-release fastener 320 can include a release 326. The release 326 is operably connected to the catch 324 to allow the catch 324 to move from the latched

position to the unlatched position when the release 326 is activated. Here, the release 326 is a button located on an upper surface of the quick-release fastener 320. The release 326 can be activated by a user pressing the button, which is connected to the catch 324 to allow the catch to retract.

[0053] The quick-release fasteners 320 provide a secure connection, but allow for easy, tool-free engagement or disengagement to remove the adjustable equipment storage rack 200 when not in use. While the quick-release fastener 320 has been discussed herein as a button-activated spring detent quick-release fastener, other quick-release fasteners can be used, such as, for example, a lever-based quick-release fastener, a cam lock quick-release fastener, or a ratchet quick-release fastener. A lever-based quick-release fastener can use a lever that operates a cam or spring-loaded mechanism to lock and release components. The lever, when activated, either tightens or loosens the fastener, allowing the components to be securely attached or easily detached without the need for tools. The cam lock quick-release fastener utilizes a rotating cam mechanism to lock or unlock components. The cam can include a slanted surface that, when rotated, such as by a knob or handle, either pulls the components together or pushes them apart. The ratchet quick-release fastener utilizes a ratcheting system to incrementally lock parts into place. The ratchet mechanism features a set of teeth that engage with corresponding teeth or gears to secure the components. The fastener is locked by rotating or pulling the mechanism until the teeth are engaged and the release occurs when the ratchet system is disengaged, usually by pulling a lever or pressing a button to release the teeth.

[0054] The quick-release fastener 320 also includes a shank 330. The shank 330 is the portion of the quick-release fastener 320 that is positioned in and slides within the slot 272 (FIG. 5) of the mounting bracket 214 (FIG. 5). The shank 330 is located between an upper flange 332 and a lower flange 334. The upper flange 332 and the lower flange 334 are located on either side of the mounting bracket plate 262 and pad 266, if used, as shown in FIGS. 8A to 8D. The upper flange 332 and the lower flange 334 prevent the quick-release fastener 320 from being removed from the slot 272 when the position of the first cradle assembly 201 is being adjusted, as will be discussed further below.

[0055] The upper flange 332 includes one or more projections. The upper flange 332 shown in FIG. 7B includes a first projection 336 and a second projection 338. The first projection 336 and the second projection 338 extend downward from the upper flange 332. The first projection 336 is engageable with the first opening 274x (FIG. 5) of each pair of openings 278 (FIG. 5) and the second projection 338 is engageable with the second opening 276x (FIG. 5) of each pair of openings 278.

[0056] FIGS. 8A to 8D illustrate the process of adjusting the position of the first cradle assembly 201. The cradle 212 is positionable in a plurality of cradle positions. At each different position of the plurality of cradle positions, the cradle 212 is a different distance from the barrier 160. Each engagement position 224 of the plurality of engagement positions 224 corresponds to one cradle position of the plurality of cradle positions, and engaging the quick-release fastener 320 with the mounting bracket 214 at one of the plurality of engagement positions 224 locates the cradle 212 at a corresponding one of the plurality of cradle positions. To move the cradle 212 from one cradle position to another, the

quick-release fastener 320 is moved from one engagement position 224 to another. As will be illustrated in FIGS. 8A to 8D, for example, the quick-release fastener 320 will be moved from the second engagement position 224b to the fourth engagement position 224d.

[0057] FIG. 8A shows the mounting bracket 214 with the quick-release fastener 320 engaged with the mounting bracket 214 in the second engagement position 224b, such as when the adjustable equipment storage rack 200 is positioned as depicted in FIG. 3A. Here, the fastener shaft 322 is located in the receiver bore 312 (FIG. 6) with the catch 324 in the latched position to secure the quick-release fastener 320 and the mounting bracket 214 to the barrier 160. The first projection 336 is engaged with the first opening 274x (FIG. 8B) of the second pair of openings 278b and the second projection 338 is engaged with the second opening 276x of the second pair of openings 278b.

[0058] As shown in FIG. 8B, the release 326 is activated to allow the catch 324 to move from the latched position to the unlatched position and the fastener shaft 322 to be withdrawn, at least in part, from the receiver bore 312. The quick-release fastener 320 can be raised until the lower flange 334 (FIG. 7B) contacts the underside surface of the mounting bracket 214. The distance between the upper flange 332 and the lower flange 334 can be set to allow the first projection 336 and the second projection 338 to disengage from the first opening 274x and the second opening 276x, respectively, of the second pair of openings 278b.

[0059] As shown in FIG. 8C, with the quick-release fastener 320 in the position discussed above, the mounting bracket 214 can be moved relative to the quick-release fastener 320, such as by sliding. The quick-release fastener 320 slides within the slot 272 as the mounting bracket 214 moves, and the fastener shaft 322 can remain partly within the receiver bore 312. When the cradle 212 is in the desired cradle position, the quick-release fastener 320 can be reinserted fully into the receiver bore 312 and the first projection 336 and the second projection 338 engaged with the first opening 274x and the second opening 276x, respectively, of the fourth pair of openings 278d, as shown in FIG. 8D. The release 326 is then deactivated, securing the quick-release fastener 320 within the receiver 310 by the catch 324 moving to the latched position. With the quick-release fastener 320 engaged with the mounting bracket 214 in the fourth pair of openings 278d, the quick-release fastener 320 is located at the fourth engagement position 224d to position the adjustable equipment storage rack 200 as depicted in FIG. 3B.

[0060] FIGS. 9A and 9B are perspective views of another cradle assembly, referred to herein as a third cradle assembly 203, that can be used as the cradle assemblies of the adjustable equipment storage rack 200. The third cradle assembly 203 is similar to the first cradle assembly 201 discussed above and that discussion applies here. The same reference numerals are used for features of the third cradle assembly 203 that are the same or similar to the corresponding features discussed above. FIG. 9A shows a front side of the third cradle assembly 203 and FIG. 9B shows a back side of the third cradle assembly 203. The cradle 212 of the third cradle assembly 203 has a different shape than the cradle 212 of the first cradle assembly 201. In the first cradle assembly 201, discussed above, the upper end 237 of the upright leg 230 is bent, but in the third cradle assembly 203, this feature is omitted and the equipment-facing surface 232 of the

upright leg **230** continues to extend upward without a bend or curve in the direction of the mounting bracket **214**.

[0061] Each cradle assembly in the adjustable equipment storage rack **200** can be the same or different. For example, the cradles **212** of each of the cradle assemblies used in the adjustable equipment storage rack **200** can be shaped differently. For example, a cradle assembly having one shape can be the first cradle assembly **201** and another having a different shape can be the third cradle assembly **203**. Other shapes and positions can be used to accommodate different pieces of equipment.

[0062] FIGS. **10A** and **10B** are perspective views of another cradle assembly, referred to herein as a fourth cradle assembly **204**, that can be used as the cradle assemblies of the adjustable equipment storage rack **200**. The fourth cradle assembly **204** is similar to the first cradle assembly **201** discussed above and that discussion applies here. The same reference numerals are used for features of the fourth cradle assembly **204** that are the same or similar to the corresponding features discussed above. FIG. **10A** shows a front side of the fourth cradle assembly **204**, and FIG. **10B** shows a back side of the fourth cradle assembly **204**.

[0063] The fourth cradle assembly **204** includes a cradle **280** that is shaped to support a different piece of equipment, such as a buoyant water mat. Such water mats can be stored in a rolled or cylindrical configuration. Here, the cradle **280** has an arcuate shape with an equipment-facing surface **282** that is also arcuate. The cradle **280** can have an arcuate shape that extends greater than **90** degrees. The cradle **280** is attached to the upright leg **230**.

[0064] The piece of equipment **10** can be secured in the cradle **280** using a strap **284**. As noted above, different straps can be used, and the strap **284** depicted in FIGS. **10A** and **10B** is a nylon strap. The water mat can be placed against the cradle **280** and then the strap **284** can be wrapped around the water mat and secured by closing a clasp **286** of the strap **284** to connect each end of the straps **284**. The strap **284** can be positioned behind the cradle **280**, such as between the cradle **280** and the upright leg **230**, to retain the strap **284** within the fourth cradle assembly **204**.

[0065] FIG. **11** is a detail view of a clasp **286** of the strap **284**. Different clasps can be used, but the clasp **286** depicted in FIG. **11** is a side-release buckle that can have two parts: one part with a prong (or tongue) and the other with a slot, and the two parts snap together. The side-release buckle can be quickly released by pressing the sides of the buckle. The nylon strap is threaded through each part of the side-release buckle, creating a loop that allows for easy length adjustment. By pulling the free end of the strap through the buckle, the length of the strap **284** can be increased or decreased to tighten the strap **284** around the water mat and adjust for mats of different sizes.

[0066] As used herein, directional terms forward (fore), aft, inboard, and outboard have their commonly understood meaning in the art. Relative to the boat, forward is a direction toward the bow and aft is a direction toward the stern. Likewise, inboard is a direction toward the center of the boat and outboard is a direction away from it.

[0067] The terms “coupled,” “fixed,” “attached,” “connected,” and the like refer to both direct coupling, fixing, attaching, or connecting, as well as indirect coupling, fixing, attaching, or connecting, through one or more intermediate components or features, unless otherwise specified herein.

[0068] Although this invention has been described with respect to certain specific exemplary embodiments, many additional modifications and variations will be apparent to those skilled in the art in light of this disclosure. It is, therefore, to be understood that this invention may be practiced otherwise than as specifically described. Thus, the exemplary embodiments of the invention should be considered in all respects to be illustrative and not restrictive and the scope of the invention to be determined by any claims supportable by this application and the equivalents thereof, rather than by the foregoing description.

What is claimed is:

1. An adjustable equipment storage rack for a boat, the adjustable equipment storage rack comprising:
 - a cradle for supporting a piece of equipment; and
 - a mounting bracket connected to the cradle to support the cradle, the mounting bracket having a plurality of engagement positions for a fastener to engage with the mounting bracket and to secure the mounting bracket to the boat at a fastening position, wherein, at each engagement position of the plurality of engagement positions, the cradle is a different distance from the fastening position.
2. The adjustable equipment storage rack of claim 1, further comprising a strap engageable with the cradle to secure the piece of equipment to the cradle.
3. The adjustable equipment storage rack of claim 1, wherein the cradle includes a first leg and a second leg oriented transversely to the first leg and connected to the first leg at a lower portion thereof.
4. The adjustable equipment storage rack of claim 3, wherein the mounting bracket is attached to a back side surface of the first leg and projects therefrom, and an upper end of the first leg is bent in the direction that the mounting bracket projects.
5. The adjustable equipment storage rack of claim 1, wherein the mounting bracket includes a plurality of openings, and each opening corresponds to one engagement position of the plurality of engagement positions and is engageable with a projection to position the bracket at a corresponding engagement position of the plurality of engagement positions.
6. The adjustable equipment storage rack of claim 5, further comprising the fastener, the fastener being a quick-release fastener.
7. The adjustable equipment storage rack of claim 5, further comprising the fastener, wherein the mounting bracket includes a slot, and the fastener is located in the slot and is capable of sliding within the slot when the mounting bracket is moved between the plurality of engagement positions.
8. The adjustable equipment storage rack of claim 7, wherein the mounting bracket includes a plurality of openings, and each opening corresponds to one engagement position of the plurality of engagement positions, and
 - wherein the fastener includes a projection engageable with an opening of the plurality of openings to position the bracket at a corresponding engagement position of the plurality of engagement positions.
9. The adjustable equipment storage rack of claim 7, wherein the slot includes a longitudinal direction and a width direction perpendicular to the longitudinal direction and the slot is elongated in the longitudinal direction,

wherein the mounting bracket includes a plurality of pairs of openings, each pair of openings corresponds to one engagement position of the plurality of engagement positions, each pair of openings includes a first opening and a second opening, and the first opening is located on one side of the slot and the second opening is located on an opposite side of the slot in the width direction, and

wherein the fastener includes a first projection engageable with the first opening and a second projection engageable with the second opening to position the bracket at a corresponding engagement position of the plurality of engagement positions.

10. A boat comprising:

a barrier; and

an adjustable equipment storage rack mounted to the barrier, the adjustable equipment storage rack including a cradle assembly, the cradle assembly including: a cradle for supporting a piece of equipment; and a mounting bracket connecting the cradle to the barrier, the mounting bracket being positionable in a plurality of cradle positions, wherein, at each position of the plurality of cradle positions, the cradle is a different distance from the barrier.

11. The boat of claim **10**, wherein the adjustable equipment storage rack further includes a fastener engageable with the mounting bracket at a plurality of engagement positions, each engagement position of the plurality of engagement positions corresponds to one cradle position of the plurality of cradle positions, and engaging the fastener with the mounting bracket at one of the plurality of engagement positions locates the cradle at a corresponding one of the plurality of cradle positions.

12. The boat of claim **11**, wherein the barrier includes a receiver and the fastener is engageable with the receiver to connect the mounting bracket to the barrier.

13. The boat of claim **12**, wherein the receiver includes a bore, and the fastener includes a fastener shaft that engages with the receiver when located in the bore, and one of the receiver and the fastener shaft includes a detent to lock the fastener shaft in the bore.

14. The boat of claim **10**, wherein the barrier has a top surface and the mounting bracket is attached to the top surface of the barrier.

15. The boat of claim **14**, wherein the majority of the cradle is located lower than the top surface of the barrier.

16. The boat of claim **10**, wherein the barrier defines an interior of the boat and separates the interior from an exterior of the boat.

17. The boat of claim **16**, wherein the cradle is located on the exterior of the boat.

18. The boat of claim **16**, further comprising a hull, the hull including a port-side hull and a starboard-side hull, the barrier being one of the port-side hull or the starboard-side hull.

19. The boat of claim **16**, wherein the cradle assembly is a first cradle assembly and the adjustable equipment storage rack includes a plurality of cradle assemblies, including the first cradle assembly and a second cradle assembly, the second cradle assembly including:

a cradle for supporting the piece of equipment; and

a mounting bracket connecting the cradle to the barrier, the mounting bracket being positionable in a plurality of cradle positions, wherein, at each position of the plurality of cradle positions, the cradle is a different distance from the barrier.

20. The boat of claim **19**, wherein the cradle of each of the first cradle assembly and the second cradle assembly has the same shape.

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